

Final Year Project Report

Kinetic Display

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A report submitted in part fulfilment of the degree of

Bachelor of Engineering (Electrical Engineering)

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CERTIFICATE

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ABSTRACT

Nowadays, appearance of product/place is more of a concern for all kinds of businesses. Kinetic display is a digital/tech solution to this. It can be installed in any place like shopping malls, banquets, museums etc to attract the people. In kinetic display, different patterns are made with the help of python programming done with the pycharm software. In kinetic display, there are 25 units (5x5 matrix) made and each unit is made by using pulleys, stepper motor, motor drive, WeMOs, End Stop, globes, RGB LEDs etc.

In Pakistan no one has ever tried to develop a live and dynamic display like this, whereas countries like Russia, Germany etc have kinetic displays installed for number of applications like advertisement significantly and then also for decoration etc. The kinetic display that we have made is less expensive as we have used stepper motors which are way cheaper than that of displays installed already, also we designed and developed/fabricated PCBs, pulleys and the casing all by ourselves.

This thesis explains kinetic display, its applications, operation, design, and development with modern world tools like CAD, CAM, PCBs, Python, IoT, CNC machines etc.

ACKNOWLEDGEMENT

Firstly we would like to thank Almighty ALLAH for giving us such strength, knowledge, ability and opportunity to complete our project. Without HIS blessings, this project would not have been possible.

Secondly we are very thankful to our project supervisor Dr. Asim Samejo for his inspiration, idea and guidance for completion of our project as well as thesis. This work would not have been possible without the guidance, motivation and enthusiasm that he has given to us.

We are also extremely thankful to the Sir Sohail and Mr. Nadir for helping us enormously to complete our project.

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LIST OF ABBREVIATIONS

Abbreviations	Full Form
CAD	Computer- Aided Design
CAE	Computer-Aided Engineering
CAM	Computer-Aided Manufacturing
EAGLE	Easily Applicable Graphical Layout Editor
EDA	Electronic Design Automation
ESP8266	Espressif Systems
GND	Ground
IP	Internet Protocol
LED	Light Emitting Diode
PCB	Printed Circuit board
RGB	Red, Green blue
SSID	Service Set Identifier
UDP	User Datagram Protocol
Wi-Fi	Wireless Fidelity
WLAN	Wireless Local Area Network

CHAPTER - 1

INTRODUCTION

1.1 Introduction to Kinetic Display

The basic idea of kinetic display is to introduce in Pakistan to this completely new piece of decorating, marketing and more. In Pakistan there are many static decorations and marketing or advertisement posters but how about if they start moving? The idea of kinetic display is to have some possibly a responsive structure or display hanging that moves rather being static or dead and people are attract to see the mesmeric display of 3D patterns. This project is successfully implemented in various museums, malls and stores in countries like Russia, Germany and South Korea. Organizations invest a lot in marketing as it is one of the key aspects that needs to be covered effectively so we are introducing this new and amusing tech for organizations to advertise effectively and make their stores or museums look different and attractive.

In Pakistan mega malls, brands, and various organizations can make very good use of this product for decorative and marketing purposes. Moreover, in Pakistan kinetic light display can impact significant change in the trend of sophisticated decoration and advertisement.

The project includes development of hardware to create various effects that may be responsive and non-responsive so that customers are attracted effectively with different color and motion effects or objects moving in-out to create some majestic and eye-catching patterns or effects.

Project also includes the use of different software to program the hardware its movement and color schemes. Program also animates the effects in laptop that can visualize and verify before installing the model physically.

Kinetic displays can be made and installed in three different fashions, in wall, on floor, and on roof. Kinetic display in the wall is installed in Russia. It is actually the future of cinema because nowadays there are 3D movies but that are just illusion whereas kinetic display installed in the wall is not an illusion, people will not need any 3D glasses to watch the 3D display.



Fig. 1.1 Kinetic Façade of Megafaces Pavilion

The Kinetic display which is installed on the floor is located in South Korea and the display that installed in the roof is in Germany.

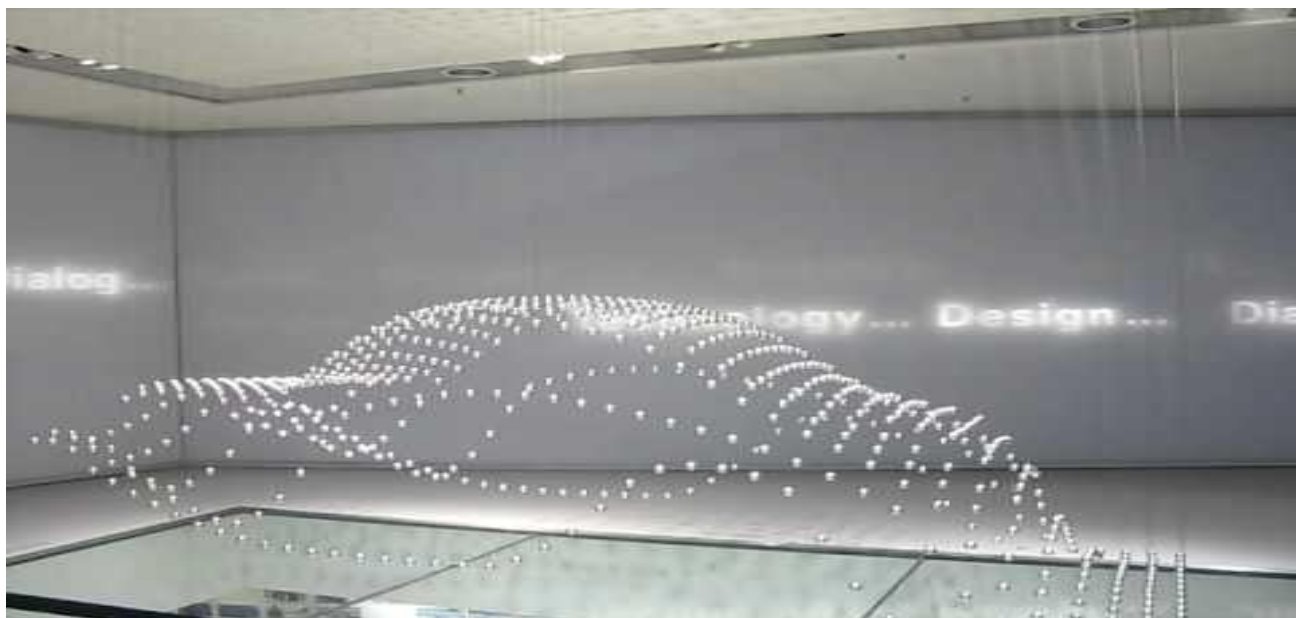


Fig. 1.2 Kinetic Display at BMW Museum



Fig. 1.3 Kinetic Pole at Hyundai Motorstudio

We proposed a 5X5 kinetic display as shown in fig. 1.2. Our kinetic display is cost effective as compared to other such displays that are installed and implemented before because here we used stepper motors instead of costly Dynamixels. One dynamixel costs Rs.6000 whereas stepper motor does the same job in less than Rs.1000. The kinetic display that we have made has an additional feature of color effects. We manipulated RGB LEDs to create eye-catching color effects.

Fig. 1.4 shows the block diagram for the system that we want to implement. First, the server or laptop sends the data bytes to the Wi-Fi microcontroller that is WeMOs and then WeMOs takes over and drives stepper motor with A4988 motor driver. WEMOS also does the job to create color effects by having control pins connected to RGB LEDs. In overall system one individual unit mainly consists of motor, motor driver, Wi-Fi built-in microcontroller, Led and more.

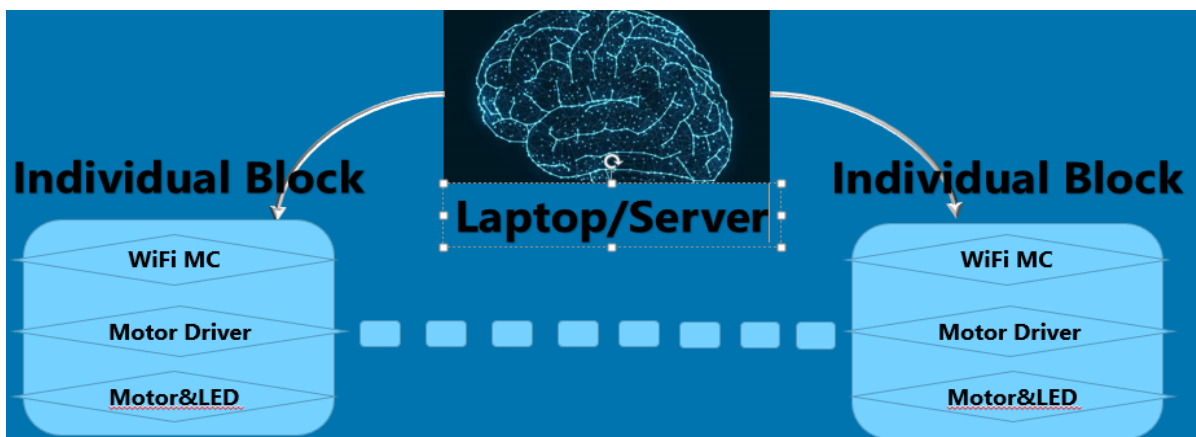


Fig. 1.4 Basic Block Diagram of Overall System

The kinetic display that we want to make patterns as shown in Fig. 1.5.

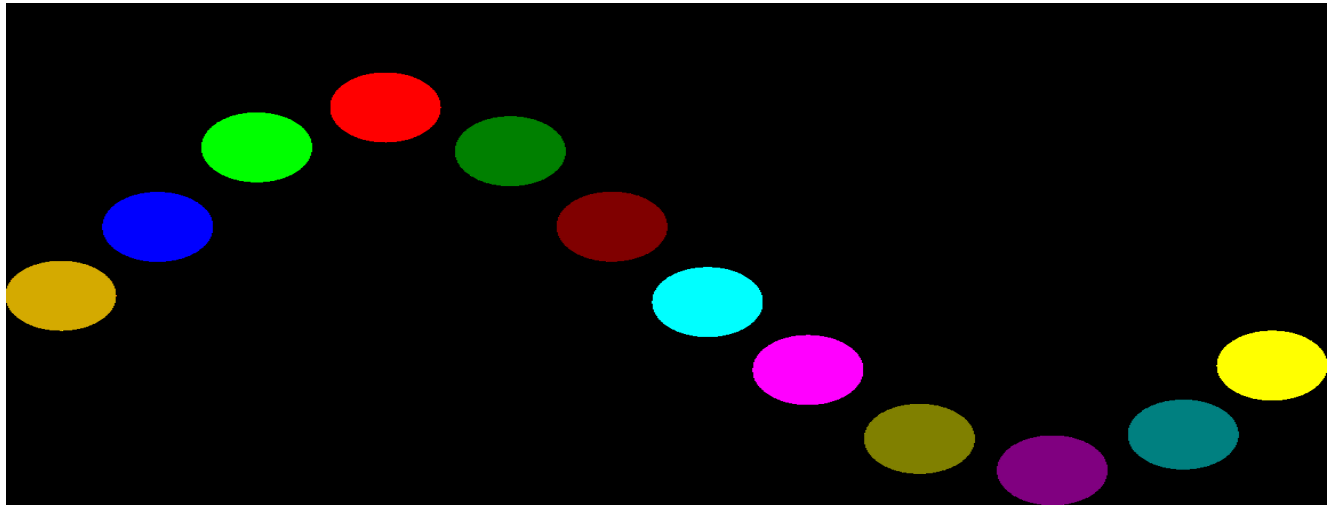


Fig. 1.5 2D Visualization of 1X12 Structure/Display

1.2 Accomplished Tasks

Through the whole process of kinetic design we have to implement these three parts which are, Electrical circuit design, Hardware design and Software implementation.

1.2.1 Electrical circuit design

In electrical circuit design we have designed these electrical circuits which are:

- PCB (printed circuit board) for the connection of WeMOs, Motor driver A4988, RGB and EndStop.
- RGB LED PCB

1.2.2 Hardware design

In hardware design, we have implemented these components

- Pulley
- 1 unit frame
- 1X5 frame
- 5X5 frame

1.2.3 Software implementation

We have many softwares like MATLAB, LabVIEW, Python etc. But we used Pycharm software by

using the python language. It is simple means can do coding in fewer steps, built-in libraries, interactive etc.

1.3 Thesis Structure

This thesis contains six chapters. The overview of the chapters are given below:

Chapter No: 2 is all about the introduction to the physical implementation, which described physical components such as WeMOs, Motor driver A4988, Pulley, Stepper motor, RGB LEDs, EndStop and Main 5X5 box.

Chapter No: 3 contains the software implementation at which the discussion about the number of softwares like MATLAB, LabVIEW and Pycharm that help us to create the code for the patterns and discussed about the libraries which have used in Pycharm software.

Chapter No: 4 discussed about the hardware design, all PCBs that were designed explained in this chapter.

Chapter No: 5 is about to assembling all the components and while assembling what problems were faced.

Chapter No: 6 contains conclusion and future recommendations.

CHAPTER - 2

PHYSICAL IMPLEMENTATION

The components that are used in kinetic display are:

- I.** Motors
- II.** WeMos
- III.** Motor driver A4988
- IV.** EndStop
- V.** RGB LEDs
- VI.** Pulley

2.1 Motors

There are different types of motors which could be used in kinetic display like DC motor, Servo motor, Stepper motor as shown in Fig. 2.1, 2.2 and 2.3.

2.1.1 DC Motor

DC motors are internally commutated DC motors (motor current direction is changed internally by a brush). These motors are designed to be used with a DC power source. DC motors consist of an armature and a permanent magnet. When current passes through the coil on the armature, a force is created due to the magnetic effect of the permanent magnet. While the armature is moving the brush of the motor changes the direction of the current mechanically, causing the shaft of the motor rotating continually [1]. By using the DC motor, we couldn't control the speed, it turns consistently when DC voltage is connected to their terminals.



Fig. 2.1 DC Motor

2.1.2 Servo Motor

Servo motors (or servos) are self-contained electric devices as shown in Fig. 2.2 that rotate or push parts of a machine with great precision. A servo motor consists of three wires a black wire connected to ground, a white or yellow wire connected to control unit and a red wire connected to power supply. The function of the servo motor is to receive a control signal that represents a desired output position of the servo shaft and apply power to its DC motor until its shaft turns to that position. It uses the position sensing device to figure out the rotational position of the shaft, so it knows which way the motor must turn to move the shaft to the instructed position [2]. As Servo motors has more precision, while using the servo motor the problem was that it was less weight bearable and also it had 180 degree rotation.



Fig. 2.2 Servo Motor

2.1.3 Stepper Motor

To see the advantages of Steeper motor such as cheap, precise accuracy of steeper motor, we move towards the Stepper motor. A “Stepper motor” or “Step motor” or “Stepping motor” is a brushless DC electric motor that works in discrete steps. It is a synchronous brushless motor where a full rotation is divided into a number of steps. The two main components of a stepper motor are the rotor and the stator. The rotor is the rotating shaft and the stator consists of electromagnets that form the stationary part of the motor. When a discrete DC voltage is applied, the stepper motor rotates in a particular angle called the step angle, thus a stepper motor is manufactured with steps per revolution of 12, 24, 72, 144, 180 and 200, with a corresponding step angle of 30, 15, 5, 2.5, 2 and 1.8. It can be operated with or without a feedback control [3].

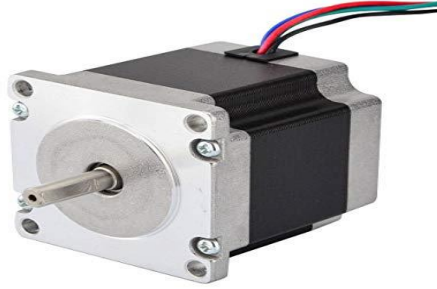


Fig. 2.3 Stepper Motor

Table 2.1 Comparison of DC Motor, Servo Motor and Stepper Motor

Sr.no	Features	DC meters	Servo motor	Stepper motor
1	Control by relay and switches	Yes	Yes	No. requirement of controller
2	Speed control	Good	Poor	Average
3	Performance in load variation	Good	Poor	Average
4	Performance in constant load	Good	Best	Good
5	Current consumption	Low	Low	High
6	Breaking system on shaft	Not available	Not available	Available
7	Position accuracy	Poor	Poor	Good
8	Speed accuracy	Good	Best	Good
9	Availability in market	Easily available	Available in selected sector	Available in selected sector
10	Driver circuit requirement for one side rotation	No	Yes	Yes

2.2 Networking Connectivity (WeMOs)

There are two kinds of networking, wired and wireless. In kinetic display, wireless network is used. Wired connectivity has a number of disadvantages such as cost, requirement of IO pins and expandability. For wireless network connection, we have a Wi-Fi controller called WeMOs-D1 R2. The features of ESP8266 are illustrated in table 2.2. In kinetic display, WeMOs ESP-12F of the ESP8266 family is used and is shown in Fig. 2.4. This Wi-Fi module is suitable for our application.

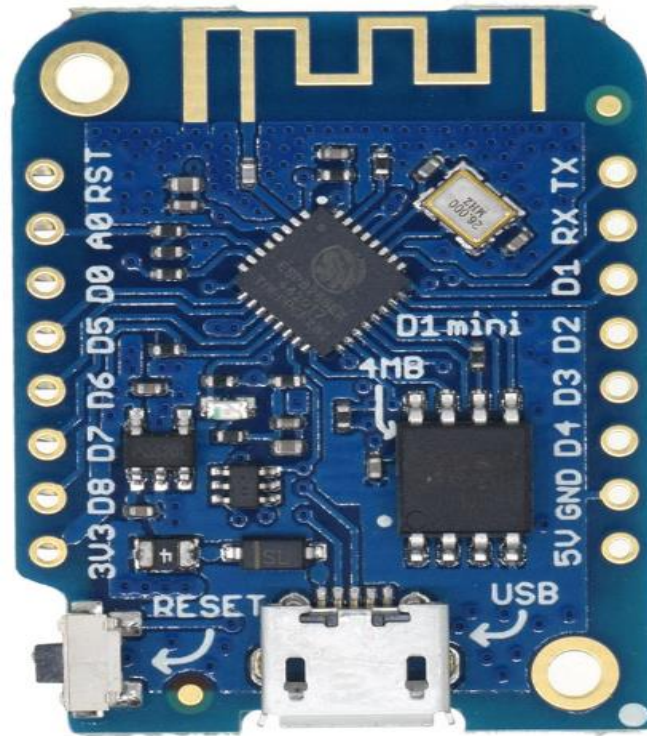


Fig. 2.2 ESP-12F

Table 2.2 ESP Families 01

Name	Active pins	Pitch	Form factor	LEDs	Antenna	Shielded	Dimensions (mm)	Notes
ESP-01	6	0.1 in	2×4 DIL	Yes	PCB trace	No	14.3 × 24.8	512 KiB Flash and blue PCB from a generic manufacturer. 1 MiB Flash, AI-Cloud and black PCB from AI-Thinker.

Name	Active pins	Pitch	Form factor	LEDs	Antenna	Shielded	Dimensions (mm)	Notes
ESP-02	6	0.1 in	2×4 castellated	No	U.FL socket	No	14.2 × 14.2	
ESP-12	14	2 mm	2×8 castellated	Yes	PCB trace	Yes	24.0 × 16.0	FCC and CE approved. ^[19]
ESP-1F	20	2 mm	2×8 castellated	Yes	PCB trace	Yes	24.0 × 16.0	FCC and CE approved. Improved antenna performance.

2.3 Pulley

In kinetic display, pulleys are designed to support the movement of bulbs up and down, and in different directions. At the first stage, we designed the pulley that was big in size i.e. 10 inches, as by using stepper motor the problem was that it had 200 steps by default per 180 degrees so the size of pulley that was designed, was 10 inches to complete a rotation of 360 degrees. While increasing the steps of stepper motor, size of pulley was reduced.



Fig. 2.3 Different Sizes of Pulley

Fig. 2.3 shows the different sizes of pulleys that were designed. The final total steps by using stepper motor were 1600 in 360 degrees of rotation which reduces the pulley size to 4.5 inches which is shown in Fig. 2.4.



Fig. 2.4 Final used pulley

2.4 Motor driver (A4988)

Generally, every motor requires specific motor driver. In table 2.1 we can see some of the motor drivers with micro-stepping for stepper motor in comparison. The A4988 is a micro stepping driver shown in Fig. 2.6 for controlling bipolar stepper motors. This implies we can control the stepper motor with only 2 pins from our controller, one controls the direction and the other controls the steps. It operates well within 10 to 30 V with peak current per phase of 2 Amp.

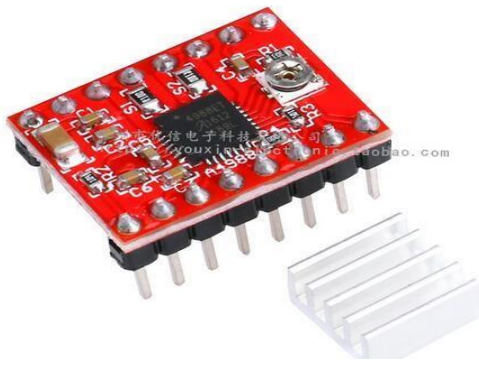


Fig. 2.5 Motor Driver (A4988)

Table 2.3 Comparison of ESP Families 02

Name	Active pins	Pitch	Form factor	LEDs	Antenna	Shielded	Dimensions (mm)	Notes
ESP-01	6	0.1 in	2×4 DIL	• Yes	PCB trace	No	14.3 × 24.8	512 KiB Flash and blue PCB from a generic manufacturer. 1 MiB Flash, AI-Cloud and black PCB from AI-Thinker.
ESP-02	6	0.1 in	2×4 castellated	No	U.FL socket	No	14.2 × 14.2	
ESP-12	14	2 mm	2×8 castellated	Yes	PCB trace	Yes	24.0 × 16.0	FCC and CE approved. ^[19]
ESP-12F	20	2 mm	2×8 castellated	Yes	PCB trace	Yes	24.0 × 16.0	FCC and CE approved. Improved antenna performance.

2.5 End-Stop

The mechanical End-Stop utilizes a switch change to identify when it is activated. The switch is wired up so when triggered, it writes the flag to LOW. End-Stop is a pressure switch that is used to tell the stepper motor about its origin where it starts its rotation.

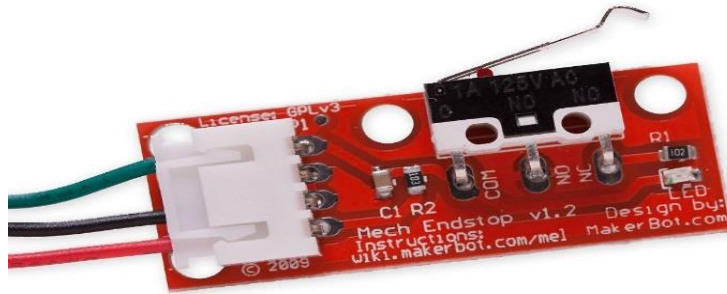


Fig. 2.6 End-Stop

2.6 RGB LEDs

RGB LED implies red, green and blue LEDs. RGB LED items consolidate these three shades to deliver more than 16 million shades of light.

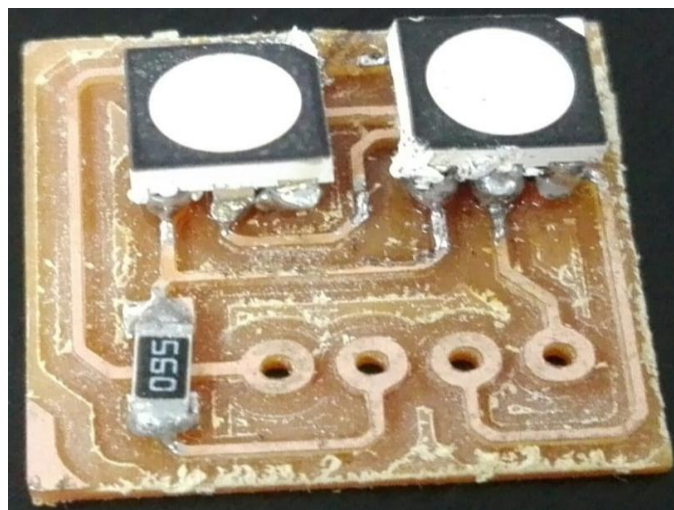


Fig. 2.7 RGB LED

The two types of RGB LEDs which are:

- Common cathode
- Common anode

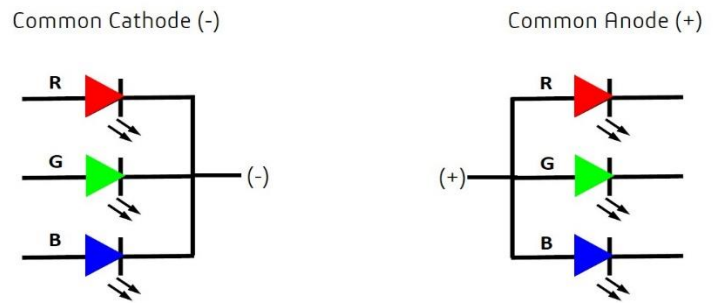


Fig. 2.8 Types of RGB LEDs

The RGB LED that is used in kinetic display is common anode.

Common Cathode: As the name suggests, is that the common cathode has all the cathodes connected together, as cathode is common, get the output from anode.

Common Anode: As the name indicates, is that the common anode has all the anodes connected together, as anode is common, get the output from cathode.

CHAPTER - 3

SOFTWARE IMPLEMENTATION

3.1 Programing Language

Several advance and compatible softwares are available to create 3D patterns such as MATLAB, LabVIEW and Pycharm, and there is a number of programming languages that can used for such 3D patterns such as Python, C/C++, JAVA, and Verilog HDL.

In MATLAB, is a proprietary software which provides simple syntax and abundant libraries for various application, whereas python is open and free, it is very easy for other parties to design packages or other software tools that extend Python.

The major difference between LabVIEW and Python is that Python was built to be a generic programming language while LabVIEW was built specially for simplified hardware integration and on the other hand python has huge array of libraries and modules for every kind of tasks.

The beauty of Pycharm software is that it supports python programming language, an advance and easy to human understandable programming language so it is the best option for kinetic display patterns, it has built-in libraries that help us to create 2D and 3D animation such as Matplotlib, NumPy, Celluloid, Timescale and Tkinter.

3.2 Libraries used from python

3.2.1 Matplotlib

The Matplotlib library is a tool and package that helps to create 2D and 3D animation, the animation tools center around the import matplotlib.pyplot as plt base class, which provides a framework around which the animation functionality is built. The main interfaces are timidAnimation and functAnimation [4].

The 5x5 matrix consist 25 objects and each object carry data for the purpose of animation.

Object is declared by assigning values to the variables such as follow.

Z1=[0 , 30 , 60 , 90 , 120] ----- upto ----- Z25 =[120,90,60,30,0]

To animate object for 3D animation, the object is created through the following function

```
def obj01():
```

```
    helix = np.vstack( ( x, a, z1 ) )
```

```
    return helix
```

and animate the object using the function.

```
Animation = [animation.FuncAnimation(fig, update_lines0, fargs=(data0, lines0), frames=3000,  
interval=100]
```

The function is self explanatory It declares the size of frame, timescale and object values.

3.2.2 NumPy

NumPy library is the package that offers wide range of functions to manipulate numerical data through operation of arithmetic such as addition, subtraction, multiplication and division as it help on creating a function to animate one individual object.

NumPy is the fundamental package for scientific computing with Python. It contains among other things.

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined. This allows NumPy to seamlessly and readily integrate with a wide variety of databases [5] such as:

```
def update_lines0(num, dataLines, lines0) :  
    for line0, data0 in zip(lines0, dataLines) :  
        line0.set_data(data0[0:2, num-1:num])  
        line0.set_3d_properties(data0[2,num-1:num])  
    return lines0
```

For 25 objects require 25 functions and the objects are animated with different data values such as:

```
lines0 = [ax.plot(data0[0][0, 0:1], data0[0][1, 0:1], data0[0][2, 0:1], 'o')[0]]
```

For 25 objects, lines must be 25 assigning values to all 25 objects.

Fig. 3.1, 3.2 and 3.3 shows the output of 3D movement patterns.

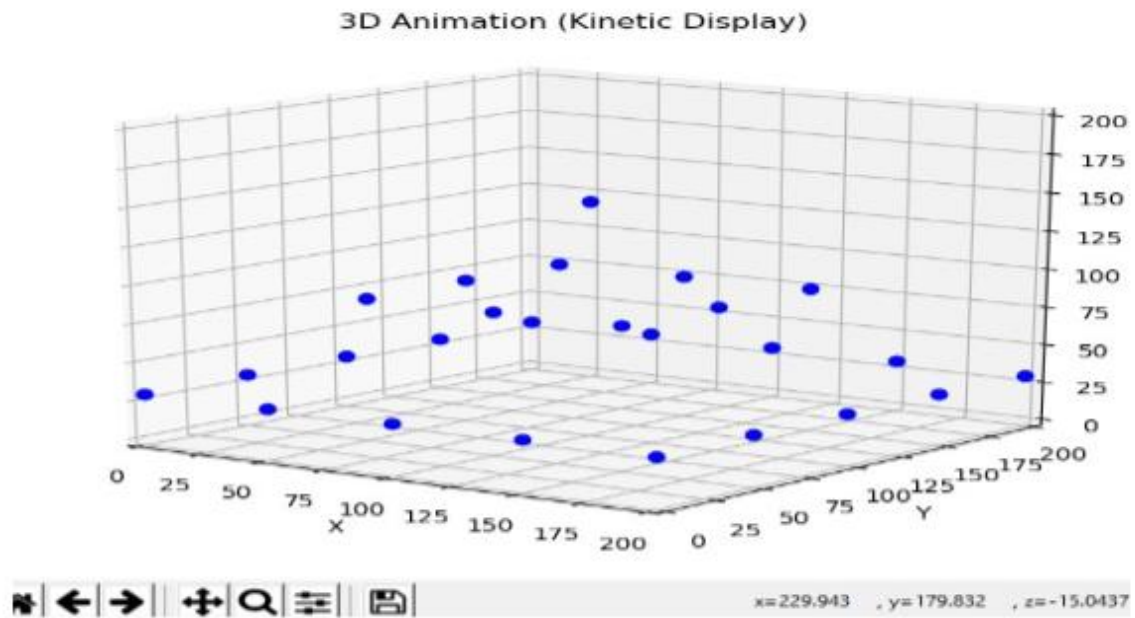


Fig. 3.1 Pattern – 1

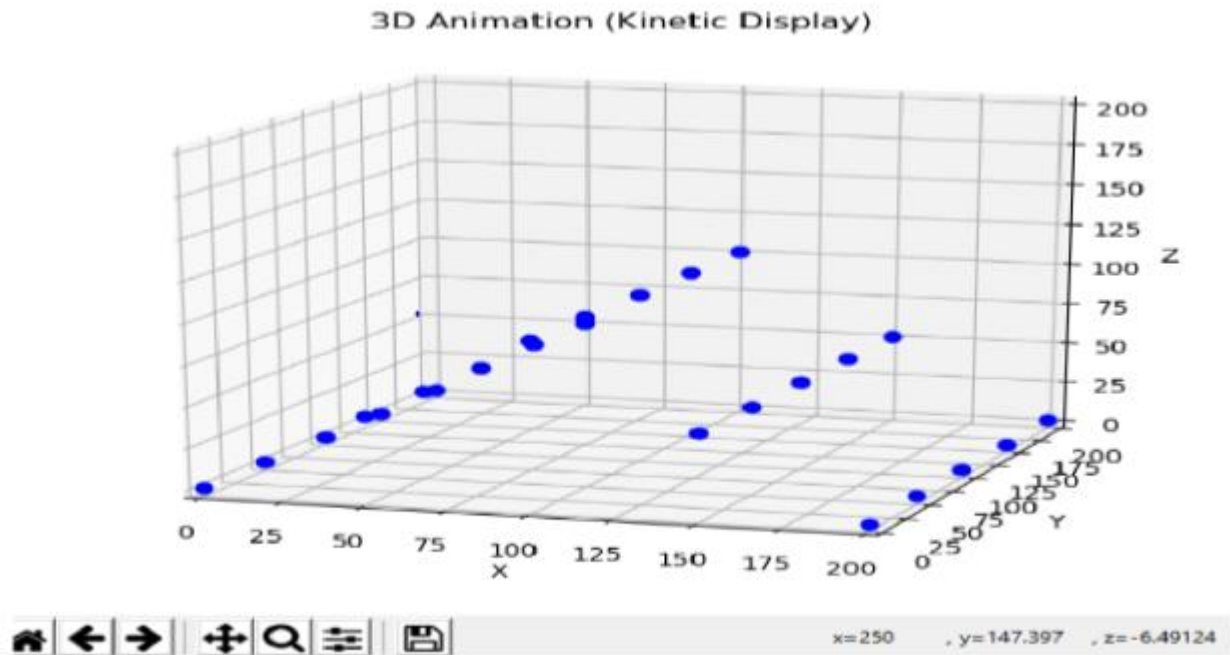


Fig. 3.2 Pattern - 2

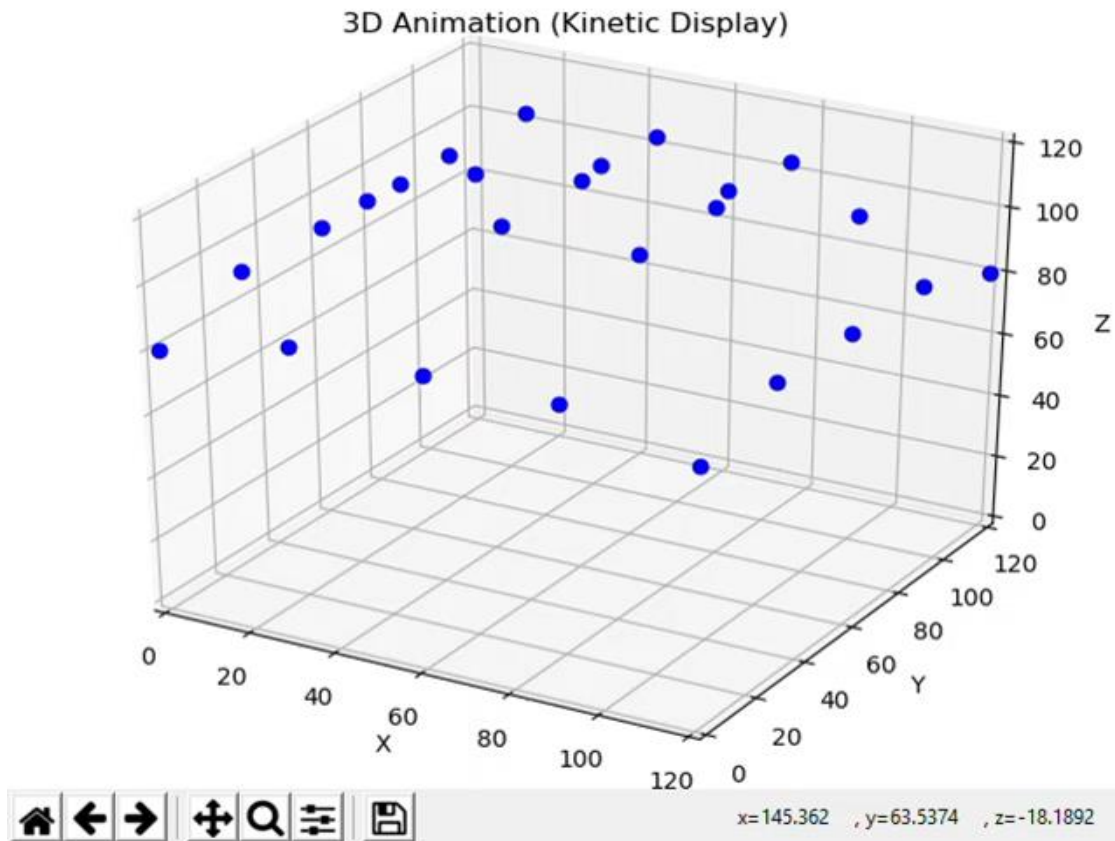


Fig. 3.3 Pattern – 3

Table 3.1 Different Patterns' Values

Number Of Nodes	Pattern-1	Pattern-2	Pattern-3
Node 0	25	10	120
Node 1	25	10	50
Node 2	25	10	50
Node 3	25	10	50
Node 4	25	10	120
Node 5	25	50	50
Node 6	75	50	100
Node 7	75	50	100
Node 8	75	50	100
Node 9	25	50	50
Node 10	25	100	50
Node 11	75	100	100
Node 12	125	100	100
Node 13	75	100	100
Node 14	25	100	50
Node 15	25	50	50
Node 16	75	50	100
Node 17	75	50	100

Node 18	75	50	100
Node 19	25	50	50
Node 20	25	10	120
Node 21	25	10	50
Node 22	25	10	50
Node 23	25	10	50
Node 24	25	10	120

In Table 3.1, rows refer to the 25 nodes and the three columns show values for three different patterns which are illustrated in Fig 3.1-3

3.2.3 Celluloid

Celluloid library consists of camera package that helps us on creating effect of 2D patterns of 7 colors,

The import function for camera is defined as,

```
from celluloid import Camera

fig = plt.figure()

camera = Camera(fig)
```

And 7 RGB Colors are defined as,

```
pallette=np.array(['red','blue','green','yellow','cyan','pink','black','white'])
```

To animate the effects of 25 objects with 7 colors for example:

```
animation = camera.animate(blit=True)
```

Fig. 3.4 shows the output of 2D animation.

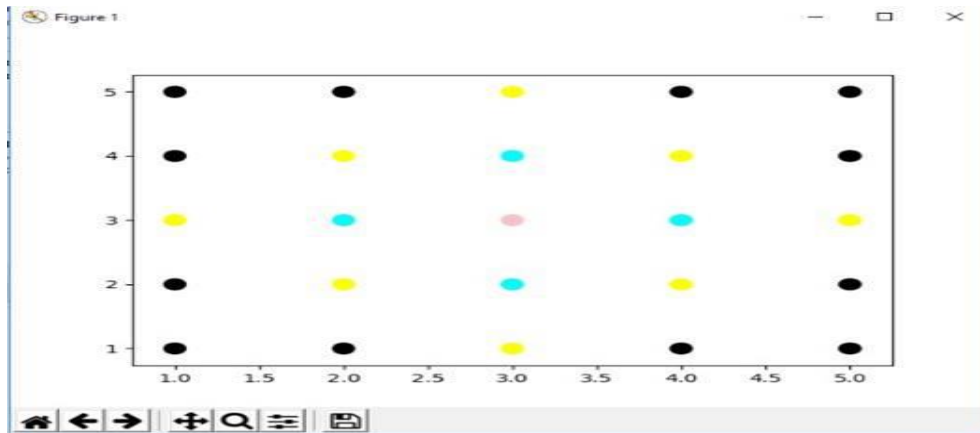


Fig. 3.4 LED Pattern

3.2.4 Timescale

Time intervals are floating-point numbers in units of seconds. Particular instants in time are expressed in seconds. There is a time module available in Python which provides functions for working with times, and for converting between representations [6].

The `sleep()` function suspends (delays) execution of the current thread for the given number of seconds. For example:

```
import time
print("This is printed immediately.")
time.sleep(2.4)
print("This is printed after 2.4 seconds.")
```

In 3D animation, the delay is required to show the shape of patterns clearly and the time library used for the delay purpose.

3.2.5 TKinter

Tkinter package is used to visualize graphics of matrix 5x5 but it does not contain the function of the movement of objects so the matplotlib is better option for 3D animation.

3.3 Networking Between Server and Client

The process of network communication is sending the information in form of bytes by using UDP protocol between server and client. In our project, the client is WeMOs and server is laptop. Laptop sends the data bytes to the Wi-Fi router and router sends them to WeMOs.

3.1.1 UDP

UDP stands for User Datagram Protocol. Data Delivery in this is not guaranteed. A datagram is a packet of information.

When using UDP, packets are just sent to the recipient. The sender will not wait for acknowledgement. It will continue sending the next packets. If

Recipient misses UDP packets the hardware may not behave as expected. If you are the recipient and you miss some UDP packets, you can't ask for those packets again. There is no guarantee you are getting all the packets and there is no way to ask for a packet again if you miss it, because of this computers can communicate more quickly.

3.4 Pseudocode of UDP Configured Client (WeMOs)

Define hardware pins

Establish connection with WAN

Write function to initiate hardware pins, wifi / network

Check if you have received any UDP packet (bytes>0)

Check packet size/length and store packet data;

Convert received packet into required type

Print received packet;

Maximum steps of stepper motor(exp:1600) = received packet ()

Set direction of stepper motor

REPEAT

3.5 Pseudocode of UDP Configured Server (Laptop)

Define whom you want to communicate with (IP, port)

Client = Define your protocol and store your destination address

Insert tuples/arrays from animation part as your data

Send data to client index/instant

REPEAT

A socket is port and in our case port is assigned to the WeMOs. Router assigns the individual IP addresses to these 25 ports or sockets.

```
address = [ ('11.11.11.11', 4211), ('11.11.11.12', 4212), ('11.11.11.13', 4213), ('11.11.11.14', 4214),  
            ('11.11.11.15', 4215),  
            (. . . . . , n(25) units)]
```

The UDP socket is setting up through the below function.

```
client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

Then the time for send packet in form of data for creating shapes like Triangle, Pyramid, tub and inverted tub to the clients such as follows:

```
Z1 = [b'20',b'131',b'-111', b'122' ,b'3' ,b'5' ,b'5' ,b'5' ,b'5']
```

```
.
```

```
.
```

```
Z25 = [b'20',b'131',b'-111', b'122' ,b'3' ,b'5' ,b'5' ,b'5' ,b'5']
```

The function for motor's movement with the use of loop is implemented by send our packet to the individual clients such as:

```
def motor():
```

```
for index in range(Len(z1)):
```

```
    client_socket.sendto(z1[index], address[0])
```

```
.
```

```
.
```

```
    client_socket.sendto(z25[index], address[24])
```

Threading Library in python is used to run multiple threads (tasks, function calls) at the same time. In that case, you probably want to look into parallel programming such as follows:

```
motorThread = threading.Thread(target = motor)
```


The confirmation of sent packets to the clients is shown below.

```
print(z1[index], 'Has been sent to ', address[0]).
```

.

.

```
print(z25[index], 'Has been sent to ', address[24])
```

CHAPTER – 4

DESIGN OF HARDWARE (NODE)

We started with testing one individual node and for that we integrated different circuit boards. We have listed them below for detailed review.

- I. Primary Driving PCB
- II. RGB LED PCB

4.1 Primary Driving PCB

The main controller PCB is connecting the controller (WeMOs) with the Motor driver (A4988), RGB LED and End Stop. Firstly, schematic is made that contains Endstop, Motor driver and RGB LED and then the main controller PCB is made through Software.

Eagle 9.2.2

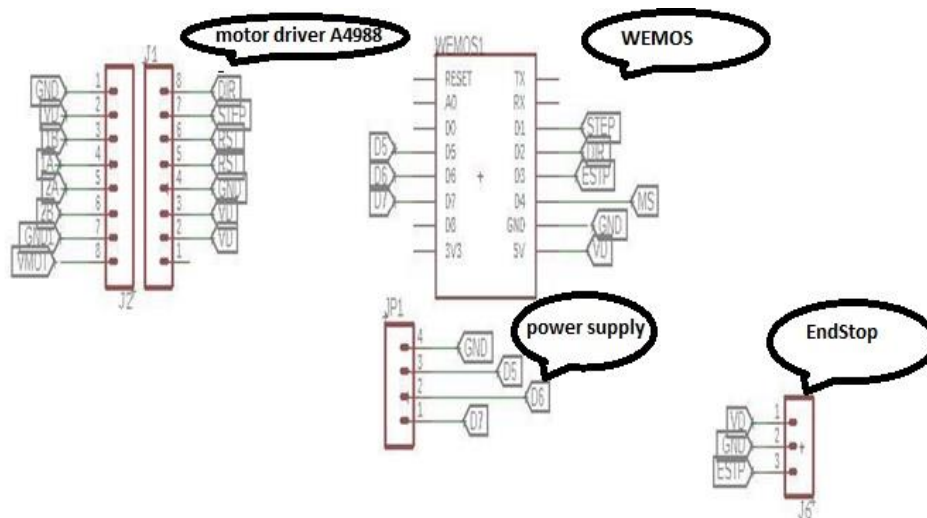


Fig. 4.1 Schematic Diagram of Main Module

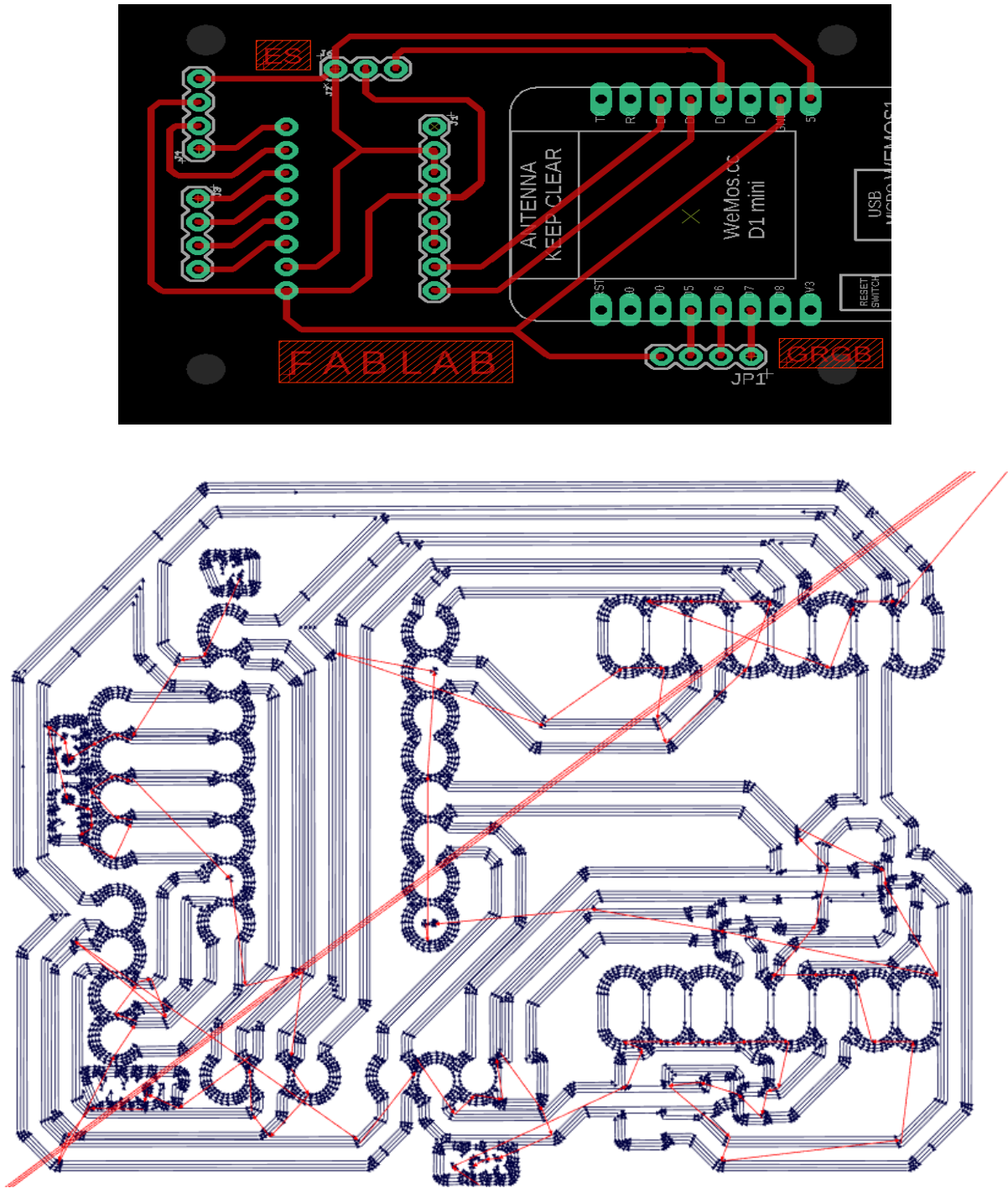


Fig 4.3 Machine Readable Files

4.2 RGB LED PCB

RGB LED File are generated in Eagle software, shown in Fig.4.3.

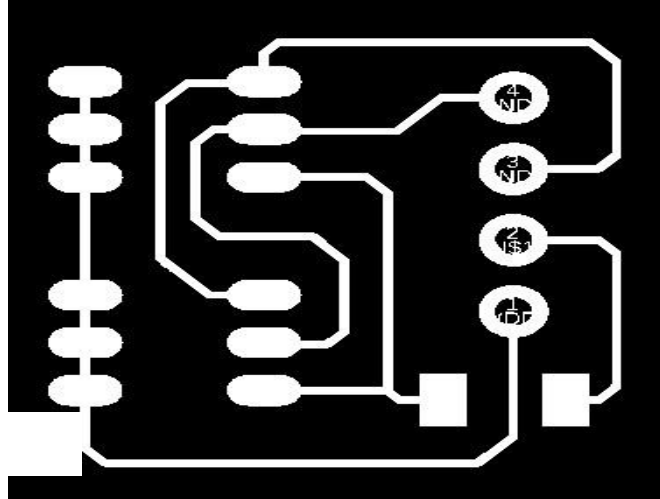


Fig. 4.4 RGB LED PCB

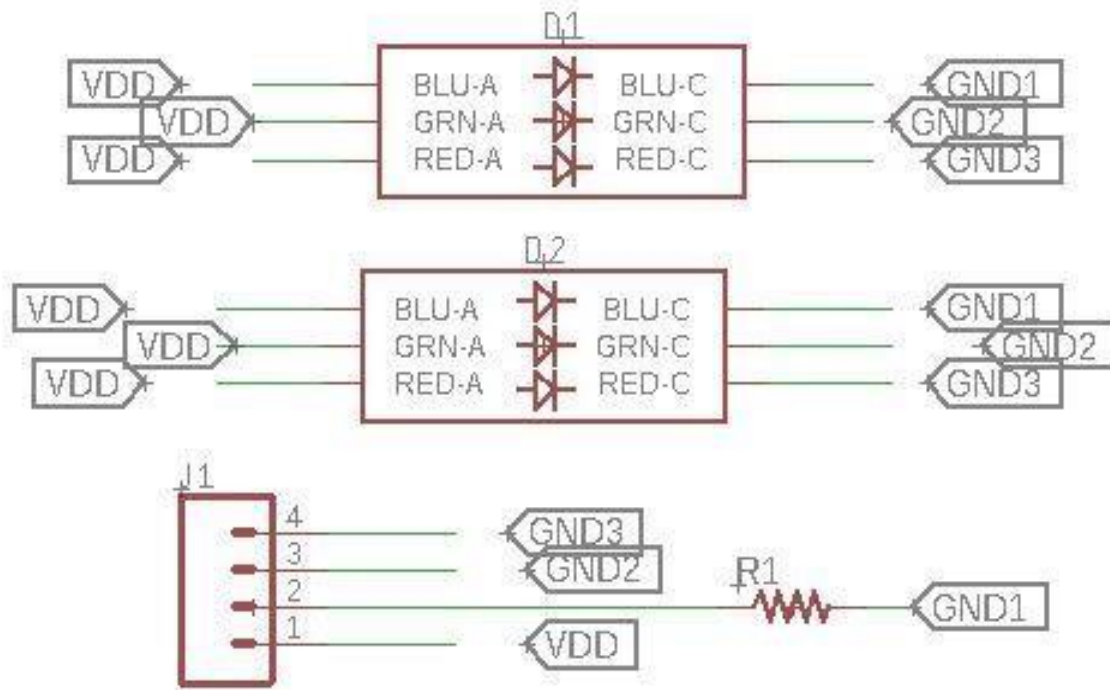


Fig. 4.5: Schematic Diagram of RGB LED

3 LEDs, red, green and blue were designed and WeMOs supplies the voltage to these LEDs. Voltage rating of red LED is 3V and voltage rating for green and blue is 5V. Initially MOSFETs were used with the WeMOs and RGB LED for more intensity in the PCB but the problem was that when MOSFET was triggered, WeMOs supplies 5V to RGB LED, as red LED's voltage rating was 3V so by getting 5V

red LED was burnt. To overcome this issue, a 56 ohm resistor was connected in series with the red LED so that red LED gets up to 3V.

4.3 Designing Hardware Using Solid Work Software

The components that are used in hardware are designed by using solid works software, solid works software is a strong displaying computer-aided design (CAD) and computer-aided engineering (CAE).

Computer program that keeps running on Microsoft Windows [7]. Here, SolidWorks 2009, version number 17 is used.

Tools which are used to design hardware are:

- I.** Sketch tools which include line, center line, arc, center circle
- II.** 3D sketch which includes extrude, dimensions.

Fig. 4.5, 4.6 and 4.7 shows the design of pulley, 1X5 box and 5X5 box. The 5X5 box, in which pulleys and all other components are placed.

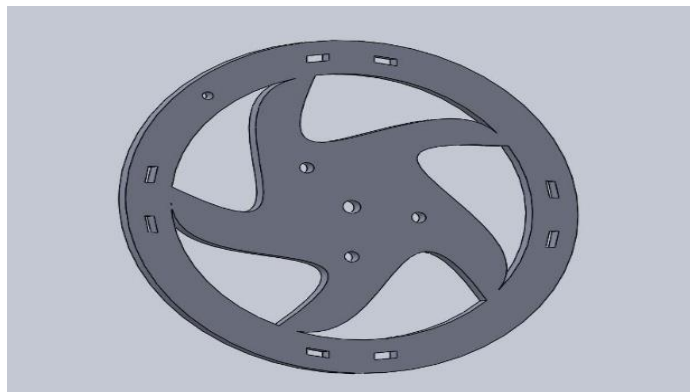


Fig. 4.5 Pulley Design

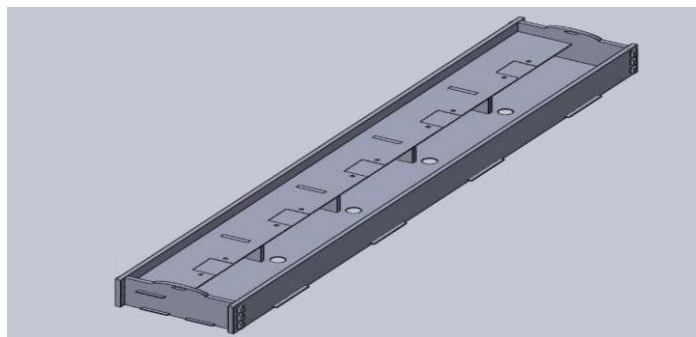


Fig. 4.6 1X5 Box Design

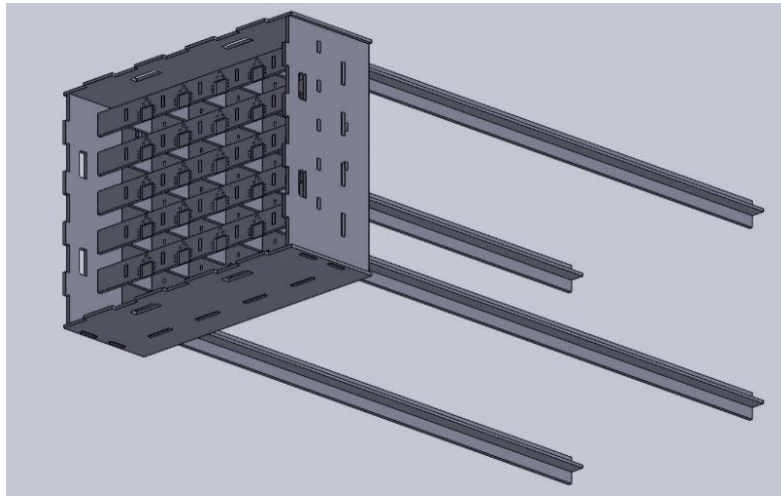


Fig. 4.7 5X5 Box Design

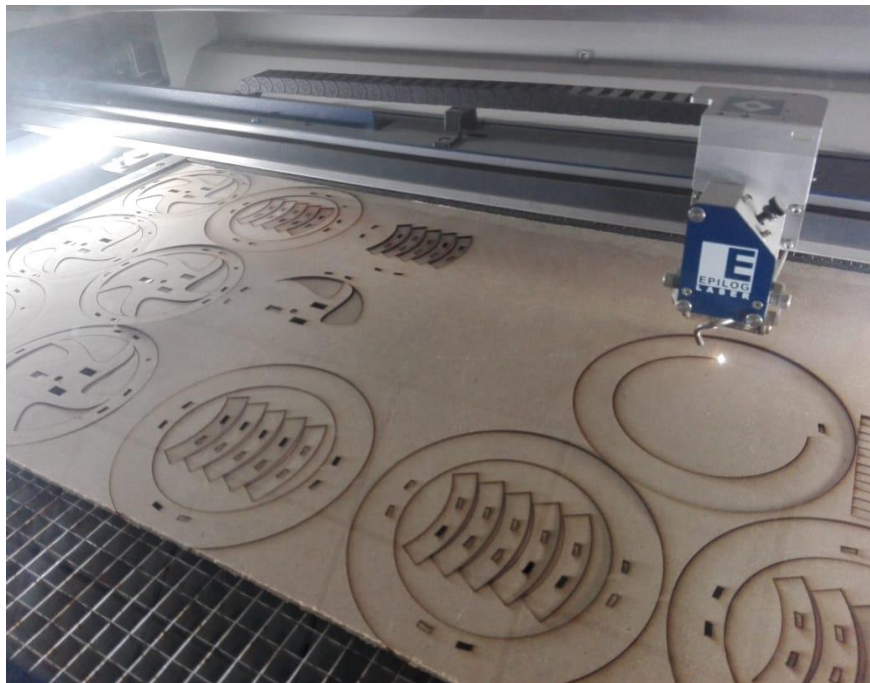


Fig. 4.8 Laser Cutting of Pulley

CHAPTER - 5

INTEGRATION OF SYSTEM

The final part is to integrate all parts of hardware to make 5X5 matrix kinetic display as the final hardware of main box of 5X5 matrix is fabricated and ready to use for kinetic display and integration of hardware it means to assemble all the components such as Stepper motors, Primary PCB, End-Stop to the main box.

5.1 Assemble Separate Box of Each Stepper Motor

At start, every stepper motor has its own designed box in order to check the working of stepper motor in box.

5.1.1 Problem

The below given Fig. 5.1 shows the box of each stepper motor. As the box of each stepper motor is designed separately so after assembling one row of each individual box of stepper motor, an intense noise came out from the overall assembling 5 boxes, this noise is not bearable that would continue running with the 5X5 kinetic display. Fig. 5.2 shows the 1 by 5 matrix display.



Fig. 5.1 Single Box of Single Stepper Motor



Fig. 5.2 1X5 Kinetic Display

5.2 Assembling Components in One Row

Fig. 5.3 shows the box of one row before assembling the components. The box of one row is designed and fabricated in order to test the response of stepper motor with the movement of RGB Led. Fig 5.3 shows the tub pattern and Fig 5.4 shows the inverted tub.



Fig. 5.3 Box of One Row



Fig. 5.4 Tub Pattern



Fig. 5.5 Inverted Tub Pattern

5.2.1 Problem

When the 5 rows designed separately and assembled together, the vibration of the noise again appears, so we move to design one complete box 5X5 matrix to reduce such errors.

5.3 Final Design 5X5 Matrix Box

Fig. 5.6 shows the 5X5 matrix box. The connection is provided through the power rail that give required voltage to every row and every module of stepper motor that is clearly shown in Fig. 5.6 and assemble all sides of the board and in Fig. 5.7 you can see the fitted stepper motors to their spaces and holding it through the white folded wire.

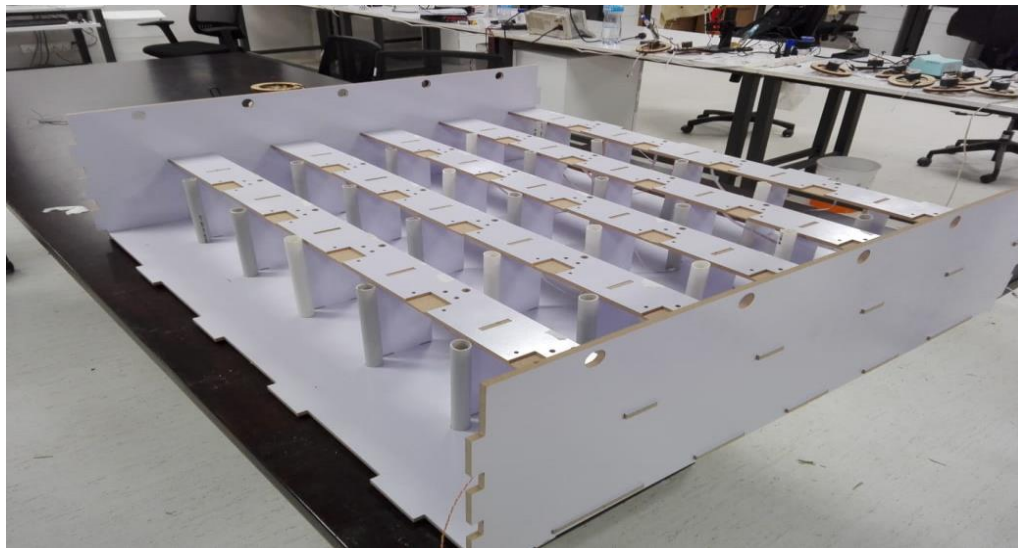


Fig. 5.6 5X5 Matrix Box



5.3.1 Problem

There is still need of mechanism which required to hold whole 5x5 matrix box up to the certain height so that public can easily observe the patterns of kinetic display.

5.4 Assembled all 25 PCBs with 5X5 Matrix Box

In Fig. 5.8 as you see, the primary PCB is placed behind every stepper motor and fixing it strongly with the box, the WeMOs, Motor driver, RGB Light and End-Stop is operating through the main PCB, as per stepper motor, required total 25 main PCBs to operate every individual stepper motor.

As all the stepper motors are connected with the box and surround with the soft foam in order to reduce the noise of the stepper motor while assembling 25 stepper motor, the small size pipe is fit into hole of the base of the main box and the RGB wire and End-Stop wire is crossing through the small size pipe so the stepper motor can easily fold and unfold the RGB wire into its back smoothly.

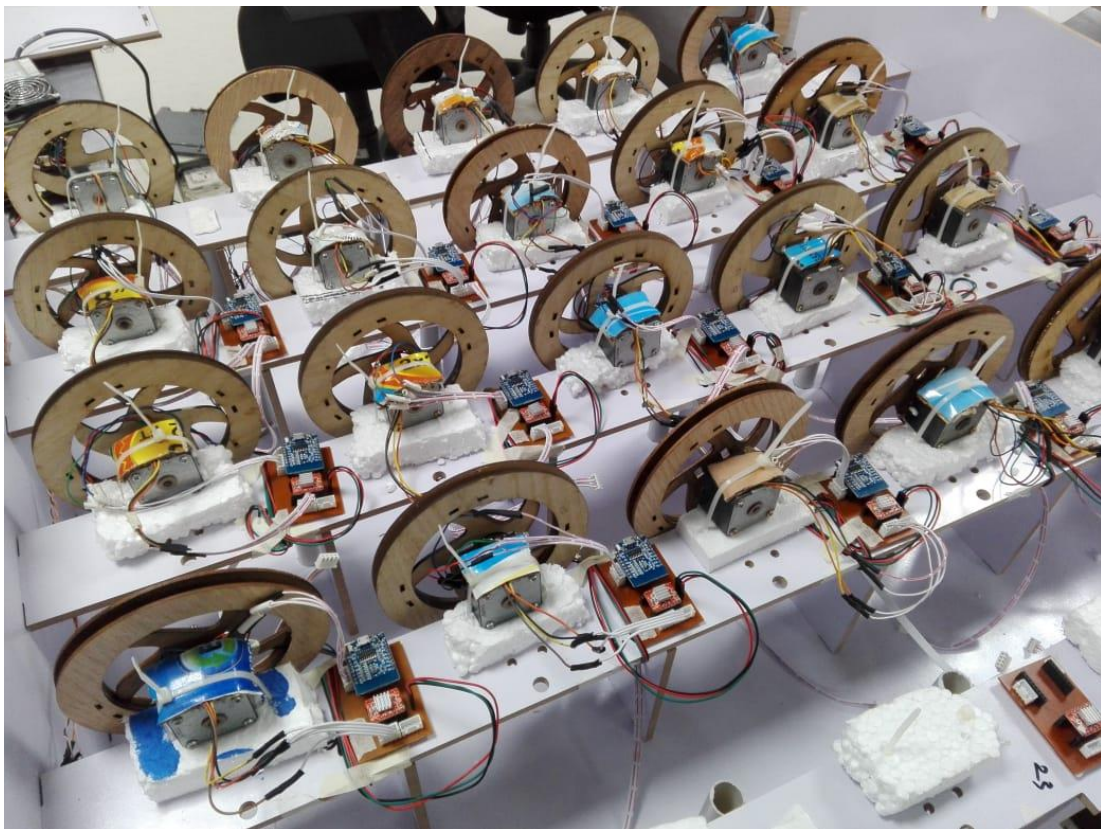


Fig. 5.8 All 25 PCBs Assembled in Box

5.5 Connection of RGB LED Savers with the Board

Fig. 5.9 shows the connection of the RGB LED savers. The RGB LEDs are connected with the LED savers and 4 very thin wires are first wound and then connected between the LED saver and the pulley which hook up with stepper motor , the colors of 4 thin wires are red, blue, green and black.



Fig. 5.9 Connection of RGB LED Bulb

There are many problems faced while making the main box of 5X5 matrix and the main problem was the dimension of the each space and holes of the box because if any single minor mistake in dimension would create big hindrance to assembling the whole box.

In order to show the output of the project to the people, the main box is put at a particular height, so that the public can easily observe which pattern is being created by 5x5 matrix kinetic display.

5.6 5X5 Kinetic Display

Finally, the 5X5 matrix kinetic display is assembled and ready to display the movement patterns and LED color patterns of the shape like tub, inverted tub, pyramid, triangle and other different patterns can be created which you want to create.

Some real time pictures of the 5X5 matrix kinetic display are shown in Fig. 5.9, Fig. 5.10 and Fig. 5.11



Fig. 5.10 Assembled Finalized Display



Fig. 5.11. While Operating 5X5 Kinetic Display



Fig. 5.12 Inverted Tub



Fig. 5.13 Random Pattern Displayed

CHAPTER - 6

CONCLUSION

By now, we have successfully made the 5×5 model of kinetic display as a prototype with RGB LEDs and motors working together to create various effects. But this is not the end. Our attempt being the first ever attempt of a Kinetic display in Pakistan had its limitations that need to be worked out, as we must see farther and farthest to make it to a commercial scale. In future, we aim to incorporate following things:

6.1 Look For More Reliable Motors

If we look at the systems commercially installed, they are very much more precise because they use very costly motors like Dynamixels of worth 45-50USD (Rs.6750-7500). Dynamixels are close loop actuators used for robotic applications with accuracy of 0.0625 degrees/step. Whereas we have used stepper motors of worth Rs.1000 only with micro-stepping of 0.225 degrees/step. Although we got this micro-stepping by reducing default 1.8 degrees/step to its 1/8th by trading off between jerk and step size. As decreasing the step size was resulting increase in jerky movement of motor. So we kind of figured out this magic number of 1/8 for our task with iterations. We are looking to use more reliable motors and explore options like servos, brushless dc motors or other. As problem with steppers is that they are not very accurate when it comes to slow motion whereas they are unmatched quick precise movement. Whereas servos are much more accurate at slow motion, hence they qualify for various relevant applications. Though mentioned options could be relatively costly, but we need to trade off the pricing with quality of effects. Also if we talk about the best options we have, it has to be close loop actuators as they will be more reliable to operate with. Hence another thing that should be considered is to have feedback so that we can track motor position every moment. This will simplify our system by eradicating end-stop (the pressure switch) we are separately using to monitor motors position. With encoders we will have much more reliable system.

6.2 Customized Globes and Robust Mechanism

Another point of concern is the density of globes/pixels. Like any other display kinetic displays also value pixels, so denser the pixels better the display. This was another problem we encountered that globes we used were cheapest (Rs.100). Hence we didn't had much options except keeping one unit far from the other so that they don't disturb each other, and this was not only because of the globes but also

the pulleys and the motor dimensions. Hence the first thing we need to do to improve our system is to add/use 3d printed globes as with 3d printing we will be able to push lengths and make denser systems also that will increase the quality of effects by making them more attractive and appealing. Whereas for pulleys we aim to use some mechanism made of gears or bearings. Next we have motors making problem, hence there are two different solutions to address this issue as following.

As we are to replace stepper motors with servos or may be brushless dc motors so that will automatically solve our issue as they are more compact in size compared to steppers.

The other thing we could do is to use one motor to control multiple units. Like if one motor controls 5 units or even more. This will also reduce processing issues.

Hence the main goal is to make our display denser as it will work more beautifully and we will be having high resolution to create more vivid patterns.

6.3 Replace Microcontroller

Replacing microcontroller is another big concern. There are following suggestions to replace microcontroller.

6.3.1 Get a dual core processor

The controller we are using currently is ESP8266 with a single core processor chip hence it is not able to process more than one task at a time. As in our application we are controlling led to create RGB effects in parallel with motor movement that results kinetic effects. Hence while working on this task we figured out this term "Multi-threading" it is a technique in writing programs for hardware and software to process multiple tasks at a time. Whereas ESP8266 does not support this feature because it has single core processor. Whereas other families in ESP and other various models like ESP12, ESP32 do have multi-core processor and are even more compact, we can also use pyboards. With rapid growth

of python language scientists and engineers have introduced micro-python to program hardware using python directly without any intermediate platform like Arduino etc. Hence there has been new hardware boards developed specially to be programmed using python, esp32 also supports micropython hence that is just another advantage to use esp32 as it can be programmed directly from python.

6.3.2 Centralize the system

Centralized system of the display will make it far more compact as there will be very less repetition of connections. There are two ideas about centralizing the system.

First is to make the whole system all n (25) units being controlled centrally using an FPGA or raspberry. With such a system we will have some limitation too but goal of centralizing the whole system will be achieved.

Other suggestion is commercially effective and sort of tradeoff between centralized and decentralized kinetic display. Idea here is to make it semi-centralized system. Like if we have 25 units so we have controller each 5 units. With this we will reduce the complexity and will have good control over too. Problem is to use/choose an effective controller other than FPGA or raspberry which is capable of driving all 5 units smoothly.

Kinetic display being a commercial project needs everything placed perfectly hence above suggestion are to address all issues we experienced to the best of our knowledge and abilities.

-

REFERENCES

- [1] <http://www.robotpark.com/Dc-Motors>
- [2] <https://www.elprocus.com/servo-motor/>
- [3] <https://www.techopedia.com/definition/13345/stepper-motor>
- [4] <https://github.com/jakevdp/PythonicPerambulations/blob/master/content/2012-08-18-matplotlib-animation-tutorial.md>
- [5] <https://www.numpy.org/>
- [6] https://www.tutorialspoint.com/python/python_date_time.htm
- [7] <https://en.wikipedia.org/wiki/SolidWorks>

COMPONENT LIST

Serial Number	Name	Price
01	Stepper Motor	1200
02	WeMOs (ESP8266)	300
03	Motor Driver (A4988)	300
04	End-Stop	200
05	RGB LED	100
06	Globe	100/300(3d printed)
07	Miscellaneous (PCB, Ply Wood/pulleys, grid, wires, power supply etc.)	1500
Total (per unit cost)		3700 / (3300+(n*300)+(n*100)+(n*200)+(n*30))

Here n is the quantity of globes per motor. You must also notice that with more globes we will need more LED's, End-Stops. Everything else will be used once.

GLOSSARY

UDP

User Datagram Protocol (UDP) is connectionless protocol. It is part of the Internet Protocol suite used by programs running on different computers on a network. UDP is used to send short messages called datagram.

IP

The Internet Protocol (IP) is the method or protocol by which data is sent from one computer to another on the Internet. Each computer (known as a host) on the Internet has at least one IP address that uniquely identifies it from all other computers on the Internet.

SSID

A service set identifier (SSID) is a sequence of characters that uniquely names a wireless local area network (WLAN). An SSID is sometimes referred to as a "network name." This name allows stations to connect to the desired network when multiple independent networks operate in the same physical area.

EAGLE

EAGLE is a scriptable electronic design automation (EDA) application with schematic capture, printed circuit board (PCB) layout, auto-router and computer-aided manufacturing (CAM) features.

WeMOs

The WeMOs D1 R2 Uno based ESP8266 a wireless 802.11 (Wi-Fi) microcontroller development board compatible with the Arduino IDE. The layout of this board is based on a standard Arduino hardware design.

CAD

CAD (computer-aided design) software is used by architects, engineers, drafters, artists, and others to create precision drawings or technical illustrations. CAD software can be used to create two-dimensional (2-D) drawings or three-dimensional (3-D) models.

CAE

CAE (computer-aided engineering) is a broad term used by the electronic design automation (EDA) industry for the use of computers to design, analyze, and manufacture products and processes.

PCB

PCB. Stands for "Printed Circuit Board." A PCB is a thin board made of fiberglass, composite epoxy, or other laminate material. Conductive pathways are etched or "printed" onto board, connecting different components on the PCB, such as transistors, resistors, and integrated circuits.

End-Stop

End-Stop is a pressure switch that is used to tell the stepper motor about its origin where it starts its rotation.

Pulley

A wheel with a grooved rim around which a cord passes, which acts to change the direction of a force applied to the cord and is used to raise heavy weights.

Electric Motor

Electric motor is the electro-mechanical machine which converts energy into mechanical energy. In other words, the devices which produce rotational force is known as the motor. The AC motor takes alternating current as an input, whereas the DC motor takes direct current.

Motor driver

Motor drives are circuits used to run a motor. In other words, they are commonly used for motor interfacing. These drive circuits can be easily interfaced with the motor and their selection depends upon the type of motor being used and their ratings (current, voltage).

RGB LED

RGB LED means red, blue and green LEDs. RGB LED products combine these three colors to produce over 16 million hues of light. Some colors are "outside" the triangle formed by the RGB LEDs. Also, pigment colors such as brown or pink are difficult, or impossible, to achieve.